

SmRF — Smart Radon Fan

Can I trust SmRF?



Smart Radon Fan

SmRF includes fault-tolerant features

- Goal: SmRF-enabled radon mitigation is as reliable as a standard radon fan installation
- SmRF builds on top of standard radon mitigation systems
- SmRF improves radon mitigation visibility



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SmRF Fault-Tolerance Summary

Failure Type	SmRF Behavior
Airthings sensor failure	Fail active (fan forced ON)
Fan control-path failure	Fail visible (user alerted)
Internet/cloud failure	Local autonomous operation continues
Home Wi-Fi/router failure	Fail visible (user alerted)
Pi IP address change	AP interface reveals new IP Local autonomous operation continues
Shelly IP address change	Automatic rediscovery and reconnection
OTA updates	Signed updates and rollback support
Power interruption	Automatic reboot and recovery



SmRF is designed so that failures degrade toward standard radon fan operation

The household loses power

- The radon fan, if active, stops as before
- When power is restored, SmRF will reboot and return to its pre-interrupted state
- The SmRF Back Office (SmRF Viewer) reports the installation as offline



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The SmRF Raspberry Pi fails

- SmRF Viewer reports the installation as offline



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SmRF loses Internet connection

- SmRF Viewer reports the installation as offline
- Locally, the SmRF continues to run, and may continue normally
- SmRF regularly and repeatedly attempts re-connection
- When connectivity is restored, SmRF re-connects automatically



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Home WiFi network fails

- SmRF Viewer reports the installation as offline
- When connectivity is restored, SmRF re-connects automatically



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Home WiFi router changes device IPs

- New SmRF Raspberry Pi IP is available from the smartphone AP interface
- SmRF Viewer is updated with the new IP
- The Shelly Smart Plug is auto-rediscovered by SmRF
- Locally, the SmRF continues to run normally



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Airthings CRM can't be read (or produces questionable readings)

- Radon fan is turned ON as default
- SmRF UI shows failure
- SmRF Viewer is informed



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Shelly Smart Plug can't be accessed

- SmRF UI shows failure
- SmRF Viewer is informed



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Worst-case scenario

- SmRF Viewer reports the installation as offline
- The Smart Plug is removed, and the fan is plugged directly into the fan receptacle



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Fault-tolerant Testing

- Development testing of identified failure modes
- In-field Beta test results



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SmRF Diagnostic AI Agent

- Available at smartradonfan.com
- Leads homeowner/installer through diagnostics and next steps
- Under development (May 2026)



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Let **SmRF**
protect your home



Know Your
System is Working



Automated
Radon Mitigation



Real-Time
Feedback & Control



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Thanks for taking a look at SmRF!